

Week 2

• Problem 1

Indexed Families of Sets and Closures

(a) Let $A_n = \{x \in \mathbb{Z} \mid |x| > n\}$ for each $n \in \mathbb{N}$. Find $\bigcap_{n \in \mathbb{N}} A_n, \bigcup_{n \in \mathbb{N}} A_n, \bigcap_{n \in \mathbb{N}} A_n^c, \bigcup_{n \in \mathbb{N}} A_n^c$.

(b) Let $\mathcal{F}$ be a family of sets such that whenever $S, T \in \mathcal{F}$, we have $S \cap T \in \mathcal{F}$. Is it always true that the intersection of all sets in $\mathcal{F}$ belongs to $\mathcal{F}$? Justify your answer.

◦ Solution 1.1

Part (a):

1. Let $x \in \mathbb{Z}$. By definition of indexed intersection, $x \in \bigcap_{n \in \mathbb{N}} A_n \iff \forall n \in \mathbb{N}, x \in A_n \iff \forall n \in \mathbb{N}, |x| > n$. Since no single integer has an absolute value strictly greater than every natural number, there is no such x . Thus:

$$\bigcap_{n \in \mathbb{N}} A_n = \emptyset$$

2. Let $x \in \mathbb{Z}$. By definition of indexed union, $x \in \bigcup_{n \in \mathbb{N}} A_n \iff \exists n \in \mathbb{N}, x \in A_n \iff \exists n \in \mathbb{N}, |x| > n$. This is true for any integer x except when $|x| \leq 1$ for $n = 1$. Specifically, for $n = 1$, $|x| > 1 \iff x \in \mathbb{Z} \setminus \{-1, 0, 1\}$. Since $A_1 \supseteq A_2 \supseteq A_3 \dots$, the largest set in the union is A_1 . Thus:

$$\bigcup_{n \in \mathbb{N}} A_n = A_1 = \{x \in \mathbb{Z} \mid |x| > 1\} = \{\dots, -3, -2, 2, 3, \dots\}$$

3. First, we find the complement $A_n^c = \{x \in \mathbb{Z} \mid |x| \leq n\}$. By definition, $x \in \bigcap_{n \in \mathbb{N}} A_n^c \iff \forall n \in \mathbb{N}, |x| \leq n$. This holds if and only if $|x| \leq 1$ (since $n \geq 1$). Thus:

$$\bigcap_{n \in \mathbb{N}} A_n^c = A_1^c = \{-1, 0, 1\}$$

4. By definition, $x \in \bigcup_{n \in \mathbb{N}} A_n^c \iff \exists n \in \mathbb{N}, |x| \leq n$. For any integer $x \in \mathbb{Z}$, we can always choose a natural number n such that $n \geq |x|$. Thus, every integer satisfies this condition:

$$\bigcup_{n \in \mathbb{N}} A_n^c = \mathbb{Z}$$

◦ Solution 1.2

Part (b):

No, it is **not always true**.

Counterexample:

Let $\mathcal{F} = \{(0, \frac{1}{n}) \mid n \in \mathbb{N}\}$, which is a family of open intervals on $\mathbb{R}$.

For any two sets $S = (0, \frac{1}{a})$ and $T = (0, \frac{1}{b})$ in $\mathcal{F}$, their intersection is $S \cap T = (0, \frac{1}{\max(a,b)}) \in \mathcal{F}$. Thus, the given condition holds.

However, the intersection of all sets in $\mathcal{F}$ is:

$$\bigcap_{n \in \mathbb{N}} (0, \frac{1}{n}) = \emptyset$$

Since the empty set $\emptyset$ is not of the form $(0, \frac{1}{n})$ for any $n \in \mathbb{N}$, $\emptyset \notin \mathcal{F}$. Therefore, the statement is false for infinite families.

• Problem 2

Equivalence Relation on Polynomials

Let $\mathbb{R}[x]$ denote the set of all polynomials with real coefficients. Define a relation $\sim$ on $\mathbb{R}[x]$ by

$$f \sim g \iff f(0) = g(0).$$

Recall that for polynomials $p, q \in \mathbb{R}[x]$, we say that p divides q , and write $p \mid q$, if there exists a polynomial $h \in \mathbb{R}[x]$ such that $q = ph$.

- (a) Prove that $\sim$ is an equivalence relation.
(b) Describe the equivalence class of the polynomial $x^2 + x + 3$.
(c) Prove that for all $f, g \in \mathbb{R}[x]$, $f \sim g \iff x \mid (f - g)$.

◦ Solution 2.1

Part (a):

- **Reflexive:** Let $f \in \mathbb{R}[x]$. Since $f(0) = f(0)$ is a basic identity, it follows directly from the definition that $f \sim f$.
- **Symmetric:** Let $f, g \in \mathbb{R}[x]$ such that $f \sim g$. By definition, this means $f(0) = g(0)$. Since equality is symmetric, $g(0) = f(0)$, which implies $g \sim f$.
- **Transitive:** Let $f, g, h \in \mathbb{R}[x]$ such that $f \sim g$ and $g \sim h$. By definition, $f(0) = g(0)$ and $g(0) = h(0)$. By transitivity of numerical equality, $f(0) = h(0)$, which implies $f \sim h$.
Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation.

◦ Solution 2.2

Part (b):

Let $p(x) = x^2 + x + 3$. Evaluating this polynomial at 0 gives $p(0) = 0^2 + 0 + 3 = 3$.

By definition, the equivalence class $[x^2 + x + 3]$ consists of all polynomials $f \in \mathbb{R}[x]$ such that $f(0) = p(0) = 3$.

Any polynomial $f(x) = a_n x^n + \dots + a_1 x + a_0$ satisfies $f(0) = a_0$. Therefore, $f(0) = 3$ means its constant term must be 3.

Thus, the equivalence class is the set of all real polynomials with a constant term equal to 3:

$$[x^2 + x + 3] = \{f \in \mathbb{R}[x] \mid f(0) = 3\} = \{x \cdot q(x) + 3 \mid q(x) \in \mathbb{R}[x]\}$$

◦ **Solution 2.3**

Part (c):

Proof:

By definition of evaluation, any polynomial $f \in \mathbb{R}[x]$ can be written uniquely as $f(x) = x \cdot q_1(x) + f(0)$ for some polynomial $q_1 \in \mathbb{R}[x]$. Similarly, $g(x) = x \cdot q_2(x) + g(0)$.

Subtracting g from f yields:

$$f(x) - g(x) = x(q_1(x) - q_2(x)) + (f(0) - g(0))$$

($\implies$) Assume $f \sim g$. By definition, $f(0) = g(0) \iff f(0) - g(0) = 0$. Substituting this back into the equation gives:

$$f(x) - g(x) = x(q_1(x) - q_2(x))$$

Since $q_1(x) - q_2(x) \in \mathbb{R}[x]$, there exists a polynomial $h = q_1 - q_2$ such that $f - g = x \cdot h$. By definition of divisibility, this means $x \mid (f - g)$.

($\impliedby$) Assume $x \mid (f - g)$. By definition of divisibility, there exists a polynomial $h \in \mathbb{R}[x]$ such that:

$$f(x) - g(x) = x \cdot h(x)$$

Evaluating both sides of this identity at $x = 0$ gives:

$$f(0) - g(0) = 0 \cdot h(0) \implies f(0) - g(0) = 0 \implies f(0) = g(0)$$

By definition of the relation, $f(0) = g(0) \iff f \sim g$.

The proof is complete.

• **Problem 3**

Degree of Polynomials and Equivalence

Let $\mathbb{R}[x] \setminus \{0\}$ denote the set of all nonzero polynomials with real coefficients. Recall that if

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0, \quad a_n \neq 0,$$

then the degree of f is $\deg f = n$, that is, the largest exponent of x with a nonzero coefficient. Define a relation $\sim$ on $\mathbb{R}[x] \setminus \{0\}$ by

$$f \sim g \iff \deg f = \deg g.$$

(a) Prove that $\sim$ is an equivalence relation.

(b) (Optional Problem) Describe the equivalence class $[f]$ of a polynomial $f \in \mathbb{R}[x] \setminus \{0\}$.

(c) (Optional Problem) On the set of equivalence classes, define

$$[f] \preceq [g] \iff \deg f \leq \deg g.$$

Prove that this relation is well-defined, i.e., that it does not depend on the choice of representatives.

(d) Prove that $\preceq$ is a partial order on the set of equivalence classes.

(e) Describe the first several equivalence classes in this order.

◦ **Solution 3.1**

Part (a):

- **Reflexive:** Let $f \in \mathbb{R}[x] \setminus \{0\}$. Since $\deg f = \deg f$ is an identity, it follows directly from the definition that $f \sim f$.
- **Symmetric:** Let $f, g \in \mathbb{R}[x] \setminus \{0\}$ such that $f \sim g$. By definition, this means $\deg f = \deg g$. Since equality is symmetric, $\deg g = \deg f$, which implies $g \sim f$.
- **Transitive:** Let $f, g, h \in \mathbb{R}[x] \setminus \{0\}$ such that $f \sim g$ and $g \sim h$. By definition, $\deg f = \deg g$ and $\deg g = \deg h$. By transitivity of numerical equality, $\deg f = \deg h$, which implies $f \sim h$. Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation.

◦ **Solution 3.2**

Part (b):

Let $f \in \mathbb{R}[x] \setminus \{0\}$ be a polynomial with $\deg f = n$.

By definition, the equivalence class $[f]$ is the set of all nonzero polynomials in $\mathbb{R}[x]$ that have the exact same degree as f :

$$[f] = \{g \in \mathbb{R}[x] \setminus \{0\} \mid \deg g = \deg f\}$$

Thus, if $\deg f = n$, $[f]$ is the set of all polynomials of degree n , which can be described as:

$$[f] = \{a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0 \mid a_i \in \mathbb{R}, a_n \neq 0\}$$

◦ **Solution 3.3**

Part (c):

Proof:

To show that the relation is well-defined, we must prove that if we choose different representatives for the same equivalence classes, the definition yields the same result.

Let $f_1, f_2, g_1, g_2 \in \mathbb{R}[x] \setminus \{0\}$ such that $[f_1] = [f_2]$ and $[g_1] = [g_2]$.

By definition of equivalence classes:

$$f_1 \sim f_2 \iff \deg f_1 = \deg f_2$$

$$g_1 \sim g_2 \iff \deg g_1 = \deg g_2$$

Now assume that $[f_1] \preceq [g_1]$ holds according to the definition, which means:

$$\deg f_1 \leq \deg g_1$$

Substituting $\deg f_1 = \deg f_2$ and $\deg g_1 = \deg g_2$ into the inequality, we get:

$$\deg f_2 \leq \deg g_2$$

By definition of the relation, this means $[f_2] \preceq [g_2]$. Thus, the truth value of the relation does not depend on the choice of representatives, meaning the relation is well-defined.

◦ **Solution 3.4**

Part (d):

Proof:

Let the set of all equivalence classes be denoted by $E = \{[f] \mid f \in \mathbb{R}[x] \setminus \{0\}\}$.

- **Reflexive:** Let $[f] \in E$. Since $\deg f \leq \deg f$ is always true for any integer $\deg f \in \mathbb{N} \cup \{0\}$, it follows by definition that $[f] \preceq [f]$.
- **Antisymmetric:** Let $[f], [g] \in E$ such that $[f] \preceq [g]$ and $[g] \preceq [f]$. By definition, this implies $\deg f \leq \deg g$ and $\deg g \leq \deg f$. By the properties of the standard order relation on integers, this means $\deg f = \deg g$. By definition of the equivalence relation, $f \sim g$, which implies $[f] = [g]$.
- **Transitive:** Let $[f], [g], [h] \in E$ such that $[f] \preceq [g]$ and $[g] \preceq [h]$. By definition, this yields $\deg f \leq \deg g$ and $\deg g \leq \deg h$. By the transitivity of numerical inequalities, $\deg f \leq \deg h$, which means $[f] \preceq [h]$. Since the relation $\preceq$ is reflexive, antisymmetric, and transitive, it is a partial order.

◦ **Solution 3.5**

Part (e):

The degree of a nonzero polynomial can be any non-negative integer $n \in \{0, 1, 2, 3, \dots\}$. The equivalence classes are ordered according to their degrees:

- The first equivalence class consists of all nonzero polynomials of degree 0 (nonzero constant polynomials):

$$[1] = \{c \in \mathbb{R} \mid c \neq 0\}$$

- The second equivalence class consists of all polynomials of degree 1 (linear polynomials):

$$[x] = \{a_1x + a_0 \mid a_1, a_0 \in \mathbb{R}, a_1 \neq 0\}$$

- The third equivalence class consists of all polynomials of degree 2 (quadratic polynomials):

$$[x^2] = \{a_2x^2 + a_1x + a_0 \mid a_2, a_1, a_0 \in \mathbb{R}, a_2 \neq 0\}$$

- The fourth equivalence class consists of all polynomials of degree 3 (cubic polynomials):

$$[x^3] = \{a_3x^3 + a_2x^2 + a_1x + a_0 \mid a_3, a_2, a_1, a_0 \in \mathbb{R}, a_3 \neq 0\}$$

Thus, the chain of the first several equivalence classes in this order is $[1] \preceq [x] \preceq [x^2] \preceq [x^3] \dots$

• **Problem 4**

Bijectivity and Inversibility of a Piecewise Function

Prove that the function $g: \mathbb{N} \rightarrow \mathbb{Z}$ defined by

$$g(n) = \begin{cases} \frac{n-1}{2}, & \text{if } n \text{ is odd,} \\ -\frac{n}{2}, & \text{if } n \text{ is even,} \end{cases}$$

is a bijection. Is it invertible? If so, find its inverse.

◦ **Solution 4.1**

- **Injective:** Let $n_1, n_2 \in \mathbb{N}$ such that $g(n_1) = g(n_2)$. We analyze the signs of the outputs:

- If n is odd, $n \geq 1 \implies n-1 \geq 0 \implies g(n) \geq 0$.
- If n is even, $n \geq 2 \implies -\frac{n}{2} \leq -1 \implies g(n) < 0$.

Since the image of odd numbers is non-negative and the image of even numbers is strictly negative, $g(n_1) = g(n_2)$ implies that n_1 and n_2 must have the same parity.

- Case 1: Both n_1, n_2 are odd.

$$\frac{n_1-1}{2} = \frac{n_2-1}{2} \implies n_1-1 = n_2-1 \implies n_1 = n_2$$

- Case 2: Both n_1, n_2 are even.

$$-\frac{n_1}{2} = -\frac{n_2}{2} \implies n_1 = n_2$$

Thus, g is injective.

- **Surjective:** Let $m \in \mathbb{Z}$ be an arbitrary integer in the codomain.

- If $m \geq 0$, let $n = 2m + 1$. Since $m \geq 0$, $n \geq 1$ and n is odd, so $n \in \mathbb{N}$. Evaluating the function gives $g(n) = \frac{(2m+1)-1}{2} = m$.
- If $m < 0$, let $n = -2m$. Since $m \leq -1$, $n \geq 2$ and n is even, so $n \in \mathbb{N}$. Evaluating the function gives $g(n) = -\frac{-2m}{2} = m$.

Thus, g is surjective.

- **Invertible and Inverse:** Since g is a bijection, it is invertible. By collecting the cases from the surjectivity proof, the inverse function $g^{-1}: \mathbb{Z} \rightarrow \mathbb{N}$ is defined by:

$$g^{-1}(m) = \begin{cases} 2m + 1, & \text{if } m \geq 0, \\ -2m, & \text{if } m < 0. \end{cases}$$

• **Problem 5**

Partitions and Preimages/Images

Suppose $f: A \rightarrow B$ is a function, $\{B_i\}_{i \in I}$ is a partition of B , and $\{A_i\}_{i \in I}$ is a partition of A . Prove that

(a) If f is surjective, then $\{f^{-1}(B_i)\}_{i \in I}$ is a partition of A .

(b) If f is injective, then $\{f(A_i)\}_{i \in I}$ is a partition of B .

◦ **Solution 5.1**

Part (a):

By definition, $\{B_i\}_{i \in I}$ being a partition of B means: (1) $B_i \neq \emptyset$ for all $i \in I$, (2) $B_i \cap B_j = \emptyset$ for all $i \neq j$, and (3) $\bigcup_{i \in I} B_i = B$. We verify these three conditions for $\{f^{-1}(B_i)\}_{i \in I}$ in A :

▪ **Non-emptiness:** Let $i \in I$. Since $\{B_i\}_{i \in I}$ is a partition, there exists some $y \in B_i$. Since f is surjective, there exists some $x \in A$ such that $f(x) = y$. By definition of preimage, $f(x) \in B_i \implies x \in f^{-1}(B_i)$. Thus, $f^{-1}(B_i) \neq \emptyset$.

▪ **Disjointness:** Let $i, j \in I$ with $i \neq j$. Suppose there exists $x \in f^{-1}(B_i) \cap f^{-1}(B_j)$. By definition of preimage, $f(x) \in B_i$ and $f(x) \in B_j$, meaning $f(x) \in B_i \cap B_j$. However, $B_i \cap B_j = \emptyset$, which is a contradiction. Thus, $f^{-1}(B_i) \cap f^{-1}(B_j) = \emptyset$.

▪ **Union:** We show $\bigcup_{i \in I} f^{-1}(B_i) = A$ by showing $x \in \bigcup_{i \in I} f^{-1}(B_i) \iff x \in A$:

$$x \in \bigcup_{i \in I} f^{-1}(B_i) \iff \exists i \in I, x \in f^{-1}(B_i) \iff \exists i \in I, f(x) \in B_i \iff f(x) \in \bigcup_{i \in I} B_i \iff f(x) \in B \iff x \in A$$

Thus, $\{f^{-1}(B_i)\}_{i \in I}$ is a partition of A .

◦ **Solution 5.2**

Part (b):

By definition, $\{A_i\}_{i \in I}$ being a partition of A means: (1) $A_i \neq \emptyset$ for all $i \in I$, (2) $A_i \cap A_j = \emptyset$ for all $i \neq j$, and (3) $\bigcup_{i \in I} A_i = A$. We verify these three conditions for $\{f(A_i)\}_{i \in I}$ in B :

▪ **Non-emptiness:** Let $i \in I$. Since $\{A_i\}_{i \in I}$ is a partition, there exists some $x \in A_i$. By definition of image, $f(x) \in f(A_i)$. Thus, $f(A_i) \neq \emptyset$.

▪ **Disjointness:** Let $i, j \in I$ with $i \neq j$. Suppose there exists $y \in f(A_i) \cap f(A_j)$. By definition of image, there exist $x_1 \in A_i$ and $x_2 \in A_j$ such that $f(x_1) = y$ and $f(x_2) = y$, meaning $f(x_1) = f(x_2)$. Since f is injective, $f(x_1) = f(x_2) \implies x_1 = x_2$. This implies $x_1 \in A_i \cap A_j$. However, $A_i \cap A_j = \emptyset$, which is a contradiction. Thus, $f(A_i) \cap f(A_j) = \emptyset$.

▪ **Union:** Note that the statement as written in the question contains a minor standard textbook assumption or typo: if $\bigcup_{i \in I} f(A_i) = f(A) = B$, it implicitly requires f to be surjective as well to partition the entire codomain B . Let us verify the union identity under the condition that it partitions $f(A)$:

$$y \in \bigcup_{i \in I} f(A_i) \iff \exists i \in I, y \in f(A_i) \iff \exists i \in I, \exists x \in A_i, f(x) = y \iff \exists x \in \bigcup_{i \in I} A_i, f(x) = y \iff \exists x \in A, f(x) = y$$

If f is a bijection (both injective and surjective), then $f(A) = B$, and thus $\{f(A_i)\}_{i \in I}$ is a partition of B .